Intelligent News Credibility Analysis & Agentic Misinformation Monitoring System

Project11 | AI/MLCourse · TeamName : LogicLeap

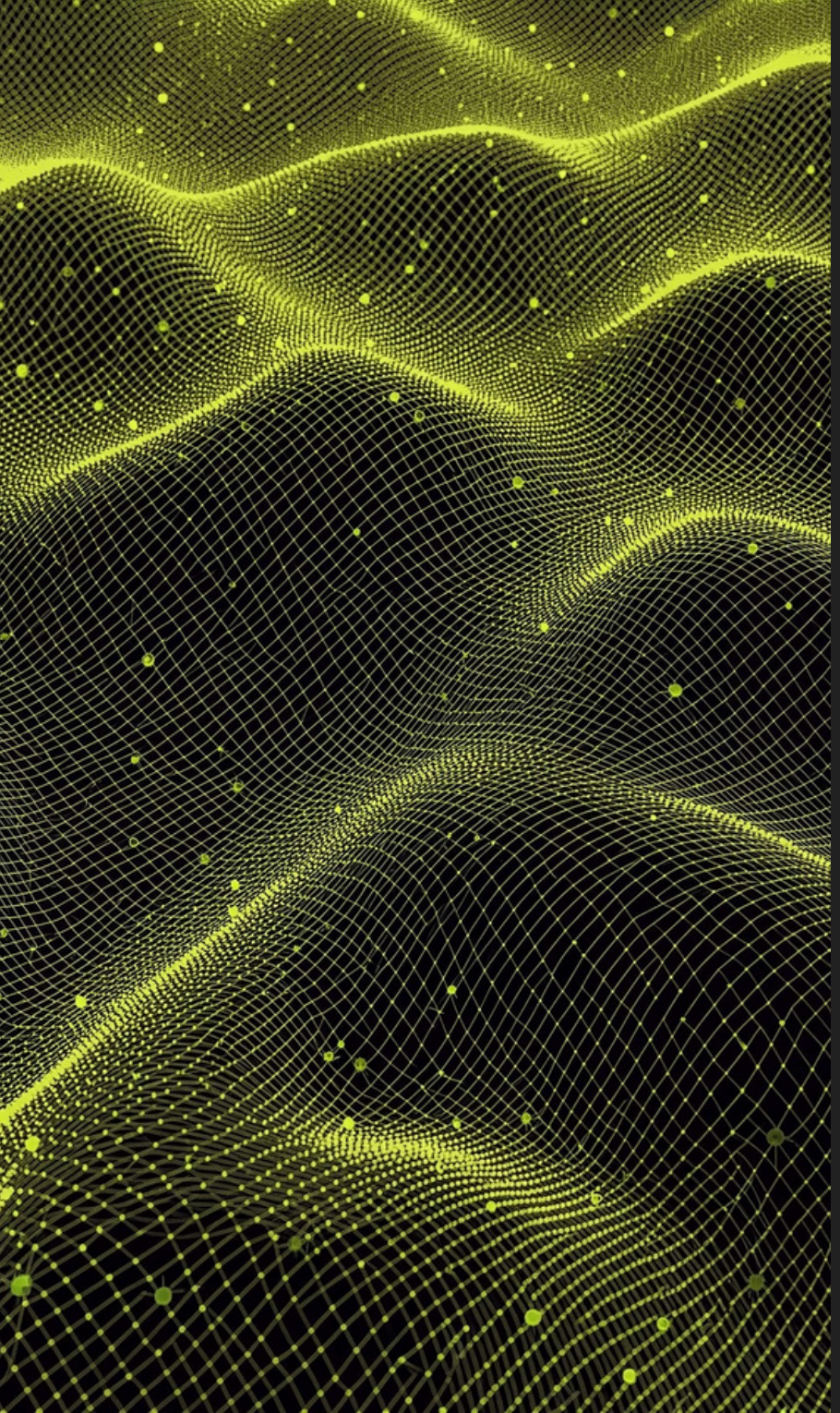
SVM · TF-IDF

LangGraph

Groq LLaMA 3

FAISS

Streamlit



Why Does This Matter?

The Problem

60%+ of people encounter fake news on social media daily

Manual fact-checking does not scale to the volume of contentonline

Existing tools are either paywalled or not explainable

What We Built

1 Milestone 1

ML classifierthatdetects fake news with confidence scores

2 Milestone 2

Agentic AI thatretrieves sources, reasons with an LLM, and generates a structured credibility report

Dataset: **WELFake** — 72,134 labelled news articles (Kaggle)

System Architecture

Orchestrated by **LangGraph StateGraph** with explicit state at each node.

Each node in the pipeline maintains explicit state, ensuring full traceability from raw article text through to the final downloadable PDF credibility report.

Milestone 1: ML Pipeline

01
Text Preprocessing
Lowercasing, URL/HTML removal, punctuation stripping, deduplication

02
Feature Extraction
TF-IDF Vectoriser: 10,000 features, bigrams, sublinear TF scaling

03
Model
LinearSVC + CalibratedClassifierCV (probability output). Rule-based override layer for strong legitimacy signals (Reuters, AP datelines)

Results

96.5% Accuracy

~96.2% Precision

~97.12% Recall

~96.6% F1 Score

These are our milestone-1 results and further for milestone -2 we have used rag with faiss for quick similarity searching and adding context, integrating llm with known datasets so that our model becomes more accurate.

Milestone 2: Agentic AI System

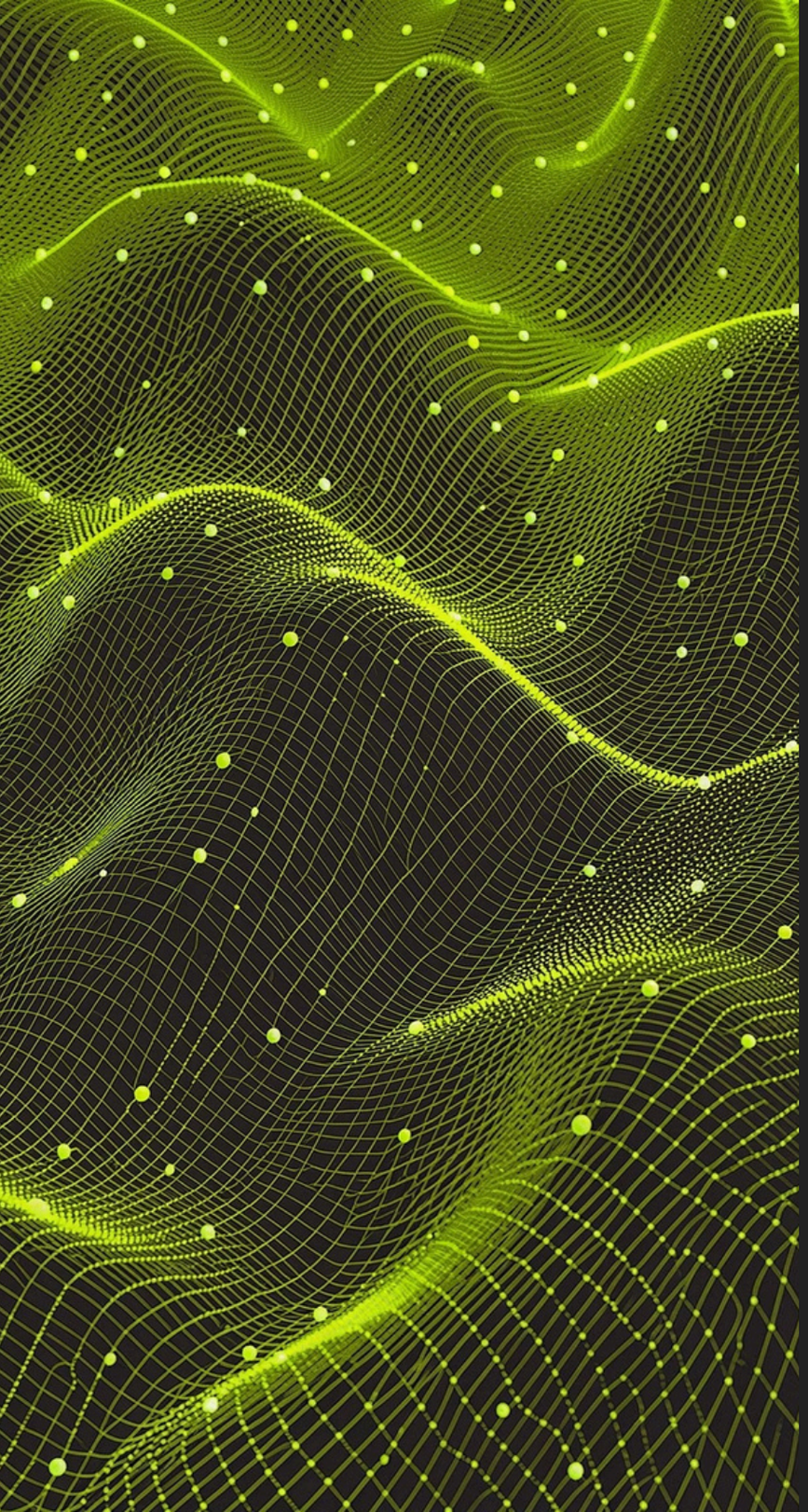
LangGraph State Machine — 4 nodes, explicit state management

Node	What it does
Predict	Runs SVM, returns label + probabilities
Retrieve	FAISS semantic search on scraped fact-check pages (Snopes, AP, PolitiFact); falls back to DuckDuckGo
Reason	Sends article + ML result + sources to Groq LLaMA 3.1 (free tier); returns structured JSON
Report	Generates a formatted PDF report using fpdf2

Output Report Includes:

- Article Summary
- Credibility Indicators
- Risk Factors
- Cross-Source Verification
- Confidence Verdict
- Ethical Disclaimer

All free-tier: Groq API (free) · FAISS (open-source) · DuckDuckGo (no API key required)



Demo & Conclusion

Live Demo

→ Input

Pasteanynews article into the Streamlit interface

→ Output

CREDIBLE / **LIKELY FAKE** / **UNCERTAIN** verdict with confidence level, retrieved sources, and downloadable PDF

Key Takeaways

Hybrid Intelligence

Combines MLprecision(SVM) with LLM reasoning (Groq LLaMA) in a single agentic pipeline

Grounded Reasoning

Retrieval-augmented generation grounds the LLM to real fact-check sources — reduces hallucination

Fully Accessible

Free-tierstack;publicly hosted on Streamlit Community Cloud